GILBERT MANGLAY

Computer Engineer | Embedded Systems, IoT & Full-Stack Automation

Cainta, Rizal, Philippines | +63 963 8964 247 | manglaygilbert18@gmail.com | github.com/ManglayGilbert

PROFESSIONAL SUMMARY

Detail-oriented Computer Engineering graduate with Latin honors (Cum Laude) specializing in hardware-software integration, IoT architecture, embedded microcontroller systems, and full-stack web platforms. Demonstrated hands-on expertise in deploying sensor networks, designing PCBs, developing Python/Django web applications, and constructing AI-driven workflow automations using n8n. Proven track record in end-user IT support, network administration, hardware troubleshooting, and system maintenance.

TECHNICAL SKILLS

- **Programming Languages:** C++, Python (Django), Java, JavaScript, Arduino C/C++, HTML5, CSS3, SQL
- **Hardware & Embedded Systems:** Microcontrollers (ESP32, Arduino, ESP8266), PCB Design, Sensor Integration, Serial Protocols (UART, I2C, SPI)
- **Software & Automation Tools:** n8n (AI Automations), KiCad, Multisim, Proteus, Arduino IDE, Git/GitHub, VS Code, PyCharm
- **Databases & Cloud Platforms:** PostgreSQL, MySQL, SQLite, Supabase, Render Deployment, REST APIs

WORK EXPERIENCE

IT Support Intern | Rentokil Initial Philippines Feb 2026 – Mar 2026

- Delivered comprehensive technical support for end-users, troubleshooting hardware, software, and network infrastructure locally and remotely.
- Installed, configured, and maintained workstations, peripheral devices, operating systems, and network peripherals across office setups.
- Managed Wi-Fi/LAN configurations, IP addressing, Active Directory task tracking, and antivirus security updates to maintain system uptime.
- Documented technical incidents, standard operational procedures, and resolution pathways to improve internal team knowledge sharing.

FEATURED ENGINEERING PROJECTS

IoT-Based Real-Time Environmental Monitoring Weather Station [ESP32, Sensors (DHT11, BMP280), Arduino IDE, Blynk IoT]

- Engineered an IoT weather monitoring unit utilizing ESP32 microcontrollers and environmental sensors to collect real-time temperature, pressure, wind velocity, and precipitation data with live web dashboard visualization.

Church Management System (CHMS) [Python, Django, PostgreSQL, JavaScript, Render]

- Architected a full-stack web application for member attendance tracking, event scheduling, and operational financial logging, deploying a PostgreSQL database backed by Render cloud infrastructure.

Button-Controlled Robot Arm with Auditory Feedback [ESP32, Stepper/Servo Motors, Digital Inputs, Arduino IDE]

- Designed and assembled an ESP32-powered robotic arm utilizing tactile button controls and digital signal processing with integrated auditory buzzer alerts for directional gripping operations.

EDUCATION

Bachelor of Science in Computer Engineering (BSCpE) Graduated 2026

Rizal Technological University | Cum Laude (GWA: 1.62)

- Relevant Coursework: Microprocessors & Microcontrollers, Embedded Systems, Digital Logic Design, Data Structures & Algorithms, Computer Networks